

raster ma scans row by row (left-to-right, top-to-bottom) many times/second and displays the stored pixel values.

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### Set-A

Attempt all the questions:

1-a) Draw the raster graphics system and explain each components of the system. [8].

Ans-

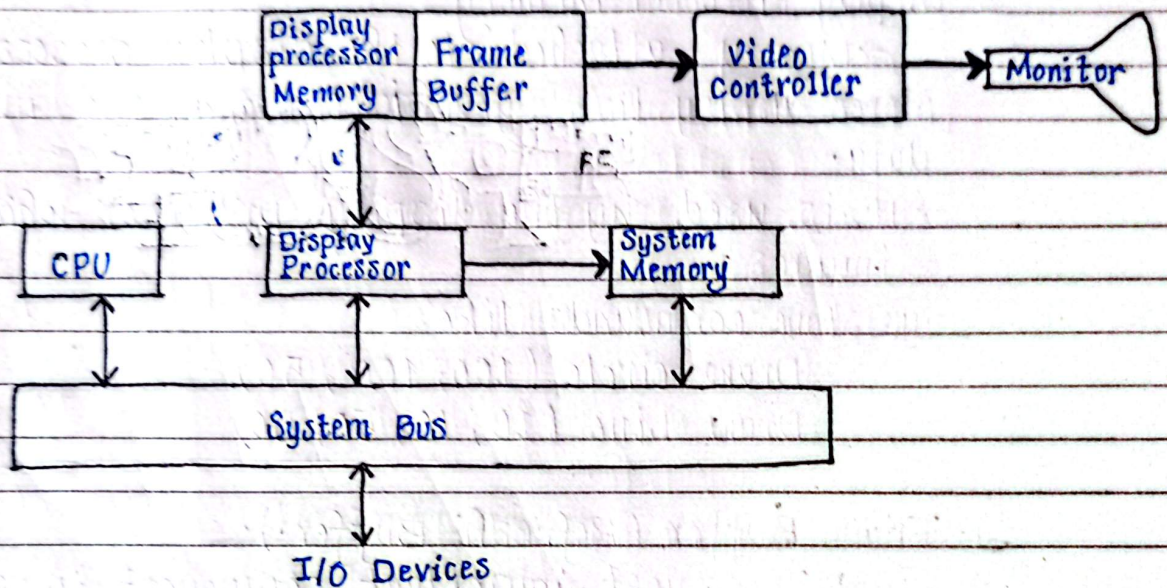


Fig: Architecture of a raster graphics system.

#### 1. CPU

It executes the application program and sends graphics commands to the display processor.

It does not directly draw pixels on the screen.

#### 2. Display Processor (Graphics Processor).

→ This is the dedicated processor for graphics operations.

It receives drawing commands from the CPU. It performs operations such as:

⊙ Line drawing

⊙ Circle generation

○ Polygon Filling

○ Transformations (rotation, scaling, translation).

It reduces the workload the CPU.

### 3. Display Processor Memory

→ Memory attached to the display processor that stores display lists, graphics command, and intermediate data.

→ It is used by the display processor while generating images.

E.g: store commands like:

Draw circle (100, 100, 50)

Draw Line (10, 10, 50, 50)

### 4. Frame Buffer (Refresh Buffer).

→ The most important component in a raster graphics system that stores the color/intensity value of every pixel on the screen.

Each memory location corresponds to one pixel.

E.g: for a 1024 x 768 display:

$$1024 \times 768 = 786,432$$

pixels must be stored in the frame buffer.

### 5. Video Controller (Refresh Controller).

It reads pixel data continuously from the frame buffer.

- It converts digital pixel values into signals required by the monitor.

It controls raster scanning and screen refreshing.

Function:

Frame Buffer  $\rightarrow$  Video Controller  $\rightarrow$  Monitor.

#### 6. Monitor (Display Device).

It displays the final image. It uses raster scanning and scans left to right, moves top to bottom and refreshes many times per time (second). (typically 60-144 Hz).

#### 7. System Memory (Main Memory).

It stores application programs and data. It is used by the CPU and display processor. It contains graphics data before it is processed.

#### 8. System Bus

It is the communication path connecting CPU, display processor, system memory, and I/O devices transferring data among all components.

#### 9. I/O Devices

Input/output devices such as keyboard, mouse, scanner, printer allowing user interaction with the graphics system.

1.6) Explain the applications of computer graphics.  
Ans- Computer graphics is the process of rendering static scene or animation on computer screen in an efficient way.

The applications of computer graphics are given below:-

### 1. Entertainment (Movies and Video Games)

Computer graphics are widely used to create visual effects, animations, characters, and realistic environments.

• Movies: CGI (Computer Generated Image) is used for special effects, animated films, and scenes that are difficult or impossible to shoot in real life.

• Video Games: Graphics create interactive 2D and 3D worlds, characters, and visual effects that enhance the gaming experience.

### 2. Graphical User Interface (GUI)

A GUI allows users to interact with computers through visual elements rather than text commands.

• Uses windows, icons, menus, buttons, and pointers.

• Makes software easier to learn and use.

E.g.: windows desktop, android apps, web browsers.

### 3. Computer-Aided Design and Manufacturing (CAD/CAM)

It is used for designing and manufacturing products accurately.

⊙ CAD: Creates detailed engineering drawings and 3D models.

⊙ CAM: Uses those designs to control manufacturing machines.

E.g: Designing cars, buildings, aircraft, machine parts.

### 4. Engineering Analysis and Business

Graphics help analyze data and present information visually.

⊙ Engineers use simulations and graphical models to test designs.

⊙ Businesses use charts, graphs, dashboards, and reports for decision-making.

E.g: Sales charts, structural stress analysis of bridges.

### 5. Medical Applications

Computer graphics assist in diagnosis, treatment, and medical education.

⊙ Visualize organs and tissues from MRI, CT, and ultrasound scans.

⊙ Create 3D models for surgical planning.

E.g: 3D heart models for surgical planning, brain scan visualization.

## 6. Computer Art

Artists use graphics software to create digital artwork and animations.

- ⊙ Supports drawing, painting, image editing and visual creativity.

E.g.: Digital paintings, illustrations, logos, posters.

## 7. Engineering Analysis

It is used to study and test engineering systems through visual simulation.

- ⊙ Helps identify design problems before actual production.
- ⊙ Reduces cost and development time.

E.g.: Aircraft airflow simulation, vehicle crash testing.

## 8. Scientific Visualization and Simulation

Helps scientists understand complex data through geographical representation.

- ⊙ Visualizes large datasets and scientific phenomena.
- ⊙ Makes analysis easier and more accurate.

E.g.: Weather forecasting, molecular structures, earthquake simulations.

## 9. Virtual Reality (VR).

Creates immersive computer-generated environments that users can explore and interact.

E.g.: VR gaming, flight simulators, virtual training systems.

2.a) Derive the Bresenham's line drawing algorithm for current decision parameter, next decision parameter and initial decision parameter when  $dx > dy$ .

Ans - Solution,

We have, initial point  $(x_0, y_0)$  and final point  $(x_n, y_n)$ .

$$dx = |x_n - x_0|$$

$$dy = |y_n - y_0|$$

Then,

As  $dx > dy$  i.e.  $|m| < 1$ .

Let  $x_n > x_0$  and  $y_n > y_0$ .

Suppose we plotted all the points from  $(x_0, y_0)$  through  $(x_k, y_k)$ .

Next Point can be either  $SE(x_{k+1}, y_k)$  or  $S(x_{k+1}, y_{k+1})$ .

Bresenham chooses the closest path.

Line can pass through either of these points.

Let the line pass through  $M(x_{k+1}, y)$

between  $SE$  and  $S$ .

$M(x_{k+1}, y)$  &

Distance between  $S(x_{k+1}, y_{k+1})$  &

is given:

$$d1 = \sqrt{((x_{k+1}) - (x_{k+1}))^2 + (y_{k+1} - y)^2}$$

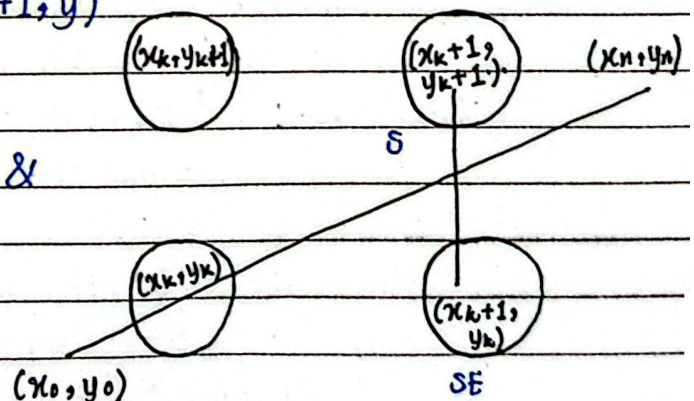
$$d1 = y_{k+1} - y$$

So, line passes through  $M(x_{k+1}, y)$

$$\therefore y = m(x_{k+1}) + c$$

Then,

$$d1 = y_{k+1} - m(x_{k+1}) - c$$



The distance between  $SE(x_{k+1}, y_k)$  and  $M(x_{k+1}, y)$  is:

$$d_2 = \sqrt{((x_{k+1}) - (x_{k+1}))^2 + (y - y_k)^2}$$

$$d_2 = y - y_k$$

$$d_2 = m(x_{k+1}) + c - y_k.$$

Then,

$$d_2 - d_1 = m(x_{k+1}) + c - y_k - y_{k-1} + m(x_{k+1}) + c.$$

$$= 2m(x_{k+1}) + 2c - 2y_k - 1.$$

$$= 2 \frac{dy}{dx} (x_{k+1}) + 2c - 2y_k - 1.$$

Eliminating complex division.

$$(d_2 - d_1) dx = 2(x_{k+1}) dy + 2c \cdot dx - 2y_k \cdot dx - dx.$$

As,  $2dy + 2c \cdot dx - dx$  is constant we discard them.

$$(d_2 - d_1) dx = 2x_k dy - 2y_k dx$$

Let  $p_k = (d_2 - d_1) dx = 2x_k dy - 2y_k dx$  be current decision parameter.

Let  $p_{k+1} = 2x_{k+1} dy - 2y_{k+1} dx$  be next decision parameter.

Since,  $dx < 0$  is not true, since  $dx = |x_n - x_0|$ .

Then, if  $p_k \geq 0$  means  $d_2 \geq d_1$ .

if  $p_k < 0$ , it means  $d_2 < d_1$ .

Then,

for  $p_k < 0$ ,  $d_2 < d_1$ . choose  $d_2$ .

This makes  $\exists E(x_{k+1}, y_k)$  closes & chooses it.

So,  $y_{k+1} = y_k$  &  $x_{k+1} = x_{k+1}$ .

Then,

$$p_{k+1} = 2(x_{k+1}) \cdot dy - 2y_k \cdot dx.$$

$$= 2x_k dy + 2dy - 2y_k \cdot dx.$$

$$\therefore p_{k+1} = p_k + 2dy.$$

for  $p_k \geq 0$  means  $d_2 > d_1$ .  
choose  $d_1$ .

$y_{k+1} = y_k + 1$  &  $x_{k+1} = x_{k+1}$ .

$$p_{k+1} = 2(x_{k+1}) dy - 2(y_k + 1) dx.$$

$$= 2x_k dy + 2dy - 2y_k dx - 2dx.$$

$$\therefore p_{k+1} = p_k + 2dy - 2dx.$$

Then,

We have,

$$p_k = (d_2 - d_1) \cdot dx$$

$$\therefore p_k = 2(x_{k+1}) dy + 2c \cdot dx - 2y_k \cdot dx - dx.$$

for final decision parameter, put  $k=0$ .

$$p_0 = (d_2 - d_1) \cdot dx$$

$$\therefore p_0 = 2(x_0 + 1) dy + 2c \cdot dx - 2y_0 dx - dx.$$

Also,

We have,

$$c = y_0 - m \left( \frac{dy}{dx} \right) x_0. \quad [y = mx + c].$$

Substituting  $\cdot$  c we have,

$$p_0 = 2(x_0 + 1) \cdot dy + 2(y_0 - \frac{dy}{dx} x_0) \cdot dx - 2y_0 dx - dx -$$

$$p_0 = 2(x_0 + 1) \cdot dy + 2dx \cdot y_0 dx - x_0 dy - 2y_0 \cdot dx - dx$$

$$p_0 = 2x_0 dy + 2dy + 2y_0 dx - 2x_0 dy - 2y_0 dx - dx$$

$$\therefore p_0 = 2dy - dx.$$

2.6) Compute all the points in the first octant for circle with center at (3,4) and radius 5.

Solution,

$$\text{Center} = (3, 4) \quad r = 5.$$

$$p = 1 - r \quad x = 0$$

$$p = 1 - 5 \quad y = r$$

$$\therefore p = -4 \quad \therefore y = 5$$

| $x = x + 1$ | if $p < 0, y = y$<br>else $y = y - 1$ | if $p < 0, p = p + 2x + 1$<br>else $p = p + 2x + 1 - 2y$ | $(x + x_c, y + y_c)$ |
|-------------|---------------------------------------|----------------------------------------------------------|----------------------|
| 0           | 5                                     | -4                                                       | (3, 9)               |
| 1           | 5                                     | -1                                                       | (4, 9)               |
| 2           | 5                                     | 4                                                        | (5, 9)               |
| 3           | 4                                     | 3                                                        | (6, 8)               |
| 4           | 3                                     | 6                                                        | (7, 7)               |

3. a) Derive the rotation matrix for a point  $P(x, y)$  rotated anti-clockwise by  $\theta$ .

Solution,

Let us rotate a point  $P(x, y)$  by angle  $\theta$  anti-clockwise with  $O(0, 0)$  being pivot point.

Let final point be  $P'(x', y')$ .

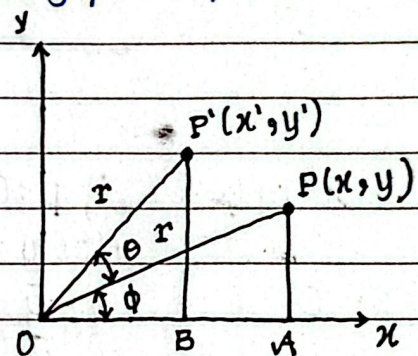
In  $\triangle OAP$ ,

$$\cos \phi = \frac{OA}{OP} = \frac{x}{r}$$

$$\text{or, } x = r \cos \phi$$

Similarly,

$$y = r \sin \phi$$



In  $\triangle OBP'$

$$\cos(\phi + \theta) = \frac{OB}{OP'} = \frac{x'}{r}$$

$$\text{or, } x' = r \cos(\phi + \theta)$$

$$\text{or, } x' = r \cos \phi \cdot \cos \theta - r \sin \phi \cdot \sin \theta$$

$$\text{or, } x' = x \cos \theta - y \sin \theta$$

$$\sin(\phi + \theta) = \frac{P'B}{OP'} = \frac{y'}{r}$$

$$\text{or, } y' = r \sin(\phi + \theta)$$

$$\text{or, } y' = r \cos \phi \cdot \sin \theta + r \sin \phi \cdot \cos \theta$$

$$\text{or, } y' = x \sin \theta + y \cos \theta$$

In matrix form.

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

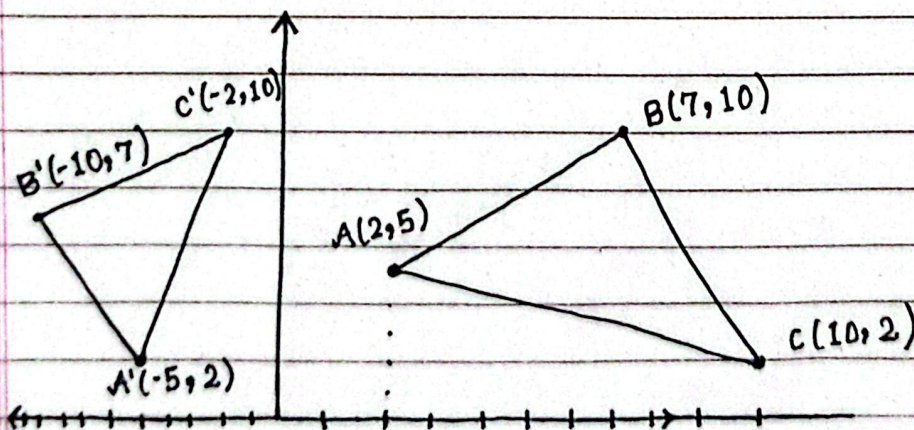
3.6) A triangle  $A(2, 5)$ ,  $B(7, 10)$  and  $C(10, 2)$  is rotated by  $90^\circ$  about origin anti-clockwise. Compute the vertices of transformed triangle. Also plot the original and transformed triangles.

Solution,

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} 2 & 7 & 10 \\ 5 & 10 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 7 & 10 \\ 5 & 10 & 2 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -5 & -10 & -2 \\ 2 & 7 & 10 \end{bmatrix}$$



4. Write short notes on:-

a) State changing Functions of OpenGL.

OpenGL is also known as the state machine.

It maintains modelview matrix, projection matrix, drawing color, background color, size of points and width of line, etc.

These state remains unchanged until we change them.

Functions of state changing include:

- `glClearColor()`
- `glColor3f()`
- `glClear()`
- `glPointSize()`
- `glLineWidth()`

the electron beam directly traces the line itself rather than scanning every pixel on the screen.

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### Set-B

1.a) Draw the vector graphics system and explain each components of the system.

Ans-

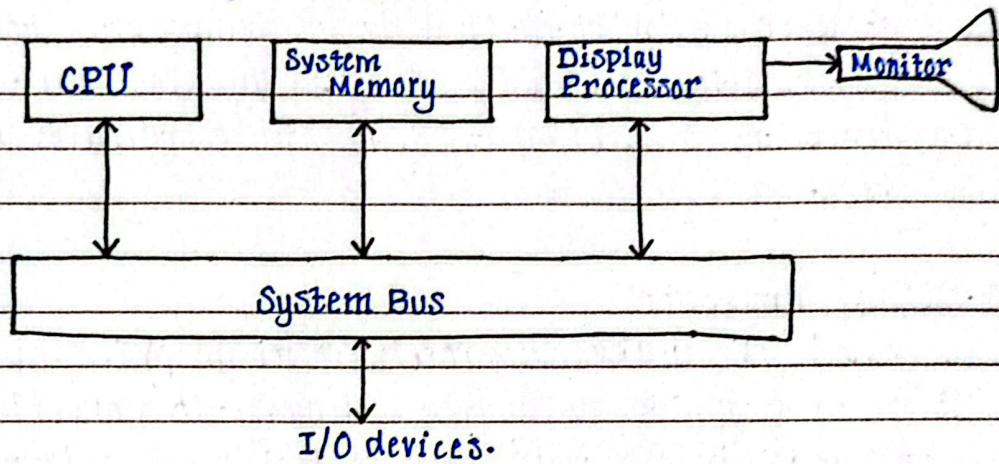


Fig: Architecture of a simple random scan system / vector graphics system.

#### 1. CPU (Central Processing Unit):

The CPU is the main controller of the system. It executes programs and processes user commands. It determines what graphics need to be displayed. It sends graphic commands (lines, curves, coordinates) to memory and the display processor.

e.g: If you draw a triangle, the CPU generates the instructions for the triangle's vertices.

## 2. System Memory

→ It stores the graphics data and application programs. It contains the coordinates of lines, curves, and other geometric primitives. It acts as temporary storage for information needed by the display processor.

e.g. for a line from  $(2, 3)$  to  $(8, 7)$ , the coordinates are stored here.

## 3. Display Processor

→ It is the dedicated processor for graphics operations. It reads graphical data from memory. It converts graphic commands into signals that control the monitor's electron beam. Offloads graphics work from the CPU, making drawing faster.

e.g. If memory stores "draw a line from A to B", the display processor calculates how the beam should move.

## 4. Monitor (Vector Display).

→ The output device that shows the image. In a vector display, the electron beam moves directly along the required lines and curves. Only the parts of the screen corresponding to the image are illuminated.

e.g. To draw a triangle, the beam traces the three sides directly instead of scanning the entire screen.

## 5. System Bus

→ The communication pathway connecting all components. It transfers data, addresses, and control signals between CPU, memory, display processor, and I/O devices.

## 6. I/O Devices

→ It allows users to interact with the system.

e.g: Keyboard, mouse, light pen, joystick, graphics tablet.

Input received from these devices is sent through the system bus to the CPU.

e.g: Moving a mouse to draw a line sends coordinate information to the CPU.

## 1.6) Explain the graphics software.

Ans- Graphics software is the collection of programs and routines that help create, edit, store, manipulate, and display graphical objects such as lines, circles, polygons, images, and animations on a computer screen.

### Functions:

- (i) Creation of Graphics.
- (ii) Editing and Manipulation.
- (iii) Viewing and display.
- (iv) Storage and retrieval.
- (v) Animation.

Graphics software is divided into two parts:

### 1. Graphics Library

A graphics library is a collection of pre-written functions that programmers use to create and manipulate graphics in a program.

Functions of a Graphics Library:

- Draw lines, circles, curves, and polygons.
- Set colors and intensity levels.
- Perform transformations such as:
  - Translation.
  - Rotation.
  - Scaling.

- Display graphical objects on the screen.

E.g: OpenGL, DirectX, Graphics.h.

Example: Instead of writing code to draw every pixel of a circle, a programmer can simply call a library function like:

circle(x, y, radius);

The graphics library handles the drawing automatically.

### 2. Graphics Application Software

Graphics application software consists of ready-made programs that allow users to create and edit graphics without programming.

### Features

- User-friendly graphical interface.
- No coding required.
- Used for drawing, designing, editing, and visualization.

E.g: Paint, AutoCAD, Adobe Photoshop, CorelDRAW.

Example: When using Paint, you can draw shapes and color them directly with tools instead of writing code.

2.a) Derive current decision parameter, next decision parameter and initial decision parameter for drawing circle with center at  $(x_c, y_c)$  and radius  $r$  using mid-point algorithm.

Ans- Solution,

Here,

Equation of a circle with center  $(x_c, y_c)$  and radius  $r$  is

$$f(x, y) = (x - x_c)^2 + (y - y_c)^2 - r^2.$$

A point lies:

- Inside the circle if  $f(x, y) < 0$ .
- On the circle if  $f(x, y) = 0$ .
- Outside the circle if  $f(x, y) > 0$ .

### ① Current Decision Parameter

Suppose the current pixel is  $(x_k, y_k)$ .

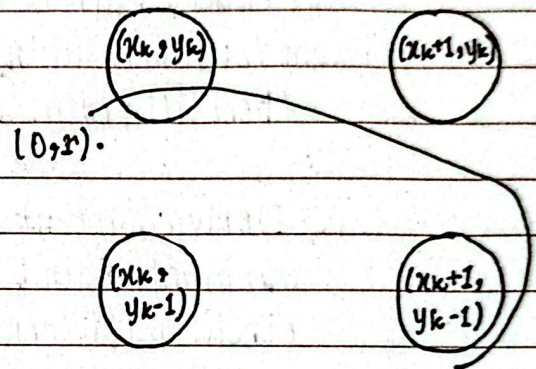
The two candidate pixels are:

Let us consider a circle of radius  $r$  and center at  $(x_c, y_c)$ . Assume that we have translated the center  $(x_c, y_c)$  to origin  $(0, 0)$ .

Assume we started to plot the circle from  $(0, r)$  to some point  $(x_k, y_k)$  on the circle as in the figure:

The next possible points are  $S(x_{k+1}, y_k)$  and  $SE(x_{k+1}, y_{k-1})$  as in figure.

Midpoint  $(M) = (x_{k+1}, y_k - \frac{1}{2})$ .



Let  $p_k = f(x_{k+1}, y_k - \frac{1}{2})$

$\therefore p_k = (x_{k+1})^2 + (y_k - \frac{1}{2})^2 - r^2$ . be the current decision parameter.

$p_{k+1} = (x_{k+1} + 1)^2 + (y_{k+1} - \frac{1}{2})^2 - r^2$ . be the next decision parameter.

if  $p_k \geq 0$ , then  $M$  lies outside, so,  $SE$  is closer to circle.

So,  $x_{k+1} = x_k + 1$  &  $y_{k+1} = y_k - 1$ .

$$p_{k+1} = (x_{k+1} + 1)^2 + (y_{k+1} - \frac{1}{2})^2 - r^2$$

$$= (x_k + 1 + 1)^2 + (y_k - 1 - \frac{1}{2})^2 - r^2$$

$$= (x_k + 1)^2 + 2(x_k + 1) + 1 + (y_k - 1/2)^2 - 2(y_k - 1/2) + 1 - r^2$$

$$= (x_{k+1})^2 + (y_k - 1/2)^2 - r^2 + 2(x_{k+1}) - 2(y_k - 1/2) + 2.$$

$$= p_k + 2(x_{k+1}) - 2(y_k - \frac{1}{2}) + 2$$

$$= \cancel{p_k + 2x_{k+1} + 2 - 2y_k - 1}$$

$$= p_k + 2x_k + 2 - 2y_k + 1 + 2$$

$$= p_k + 2x_k - 2y_k + 5.$$

Then,

$$x_{k+1} = x_k + 1, \quad y_{k+1} = y_k - 1$$

$$= p_k + 2(x_{k+1} - 1) - 2(y_{k+1} + 1) + 5$$

$$\therefore p_{k+1} = p_k + 2x_{k+1} - 2y_{k+1} + 1.$$

if  $p_k < 0$  then  $M$  lies inside, so  $S$  is closer to circle.  
So,

$$x_{k+1} = x_k + 1 \quad \& \quad y_{k+1} = y_k.$$

$$p_{k+1} = (x_{k+1} + 1)^2 + (y_{k+1} - \frac{1}{2})^2 - r^2$$

$$= (x_k + 1 + 1)^2 + (y_k - \frac{1}{2})^2 - r^2.$$

$$= (x_k + 1)^2 + 2(x_k + 1) + 1 + (y_k - \frac{1}{2})^2 - r^2$$

$$= (x_k + 1)^2 + (y_k - \frac{1}{2})^2 - r^2 + 2(x_k + 1) + 1$$

$$= p_k + 2(x_k + 1) + 1 \quad [x_{k+1} = x_k + 1.]$$

$$\therefore p_{k+1} = p_k + 2x_{k+1} + 1.$$

for initial decision parameter, we use initial point  $(0, r)$ .

$$p_0 = (x_0 + 1)^2 + (y_0 - 1/2)^2 - r^2.$$

$$p_0 = (0 + 1)^2 + (r - 1/2)^2 - r^2.$$

$$p_0 = 1 + r^2 - 2r \cdot \left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)^2 - r^2$$

$$\therefore p_0 = \frac{5}{4} - r.$$

2.6) Compute all the points on the line (3,8) and (-5,2) using Bresenham algorithm and plot the line.

Solution,

$$(x_1, y_1) = (3, 8) \text{ , } (x_2, y_2) = (-5, 2).$$

$$\begin{aligned} dx &= |x_2 - x_1| & dy &= |y_2 - y_1| \\ &= |-5 - 3| & &= |2 - 8| \\ &= 8 & &= 6 \end{aligned}$$

Since,  $dx > dy$ .

$$\text{Step} = dx = 8.$$

Since,  $x_1 > x_2$ ,  $x_{inc} = -1$

Since,  $y_1 > y_2$ ,  $y_{inc} = -1$ .

$$\begin{aligned} p &= 2dy - dx \\ &= 2 \times 6 - 8 \end{aligned}$$

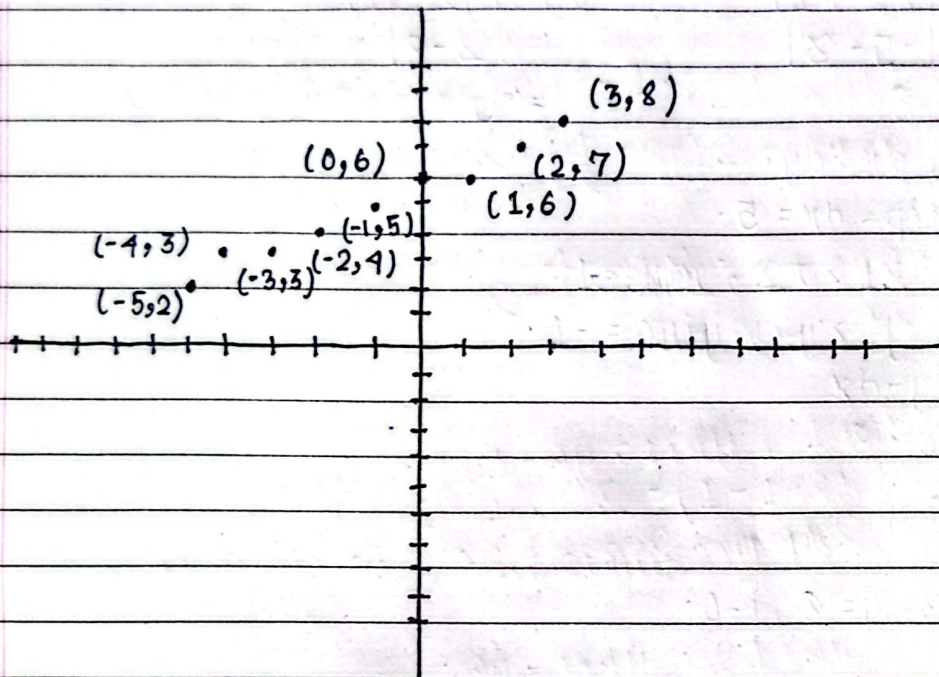
$$\therefore p = 4.$$

Then,

$$x = 3, y = 8, i = 0.$$

| i | <del>if <math>p &lt; 0, x = x</math></del> | <del><math>y += y_{inc}</math></del> | <del>if <math>p &lt; 0, p = p + 2dy</math></del> |          |
|---|--------------------------------------------|--------------------------------------|--------------------------------------------------|----------|
|   | <del>else <math>x += x_{inc}</math></del>  |                                      |                                                  |          |
| i | $x += x_{inc}$                             | if $p < 0, y = y$                    | if $p < 0, p = p + 2dy,$                         | $(x, y)$ |
|   |                                            | else $y += y_{inc}$                  | else, $p = p + 2dy - 2dx$ .                      |          |
|   | 3                                          | 8                                    | 4                                                | (3, 8)   |
| 0 | 2                                          | 7                                    | 0                                                | (2, 7)   |
| 1 | 1                                          | 6                                    | -4                                               | (1, 6)   |
| 2 | 0                                          | 6                                    | 8                                                | (0, 6)   |
| 3 | -1                                         | 5                                    | 4                                                | (-1, 5)  |
| 4 | -2                                         | 4                                    | 0                                                | (-2, 4)  |

|   |    |   |    |           |
|---|----|---|----|-----------|
| 5 | -3 | 3 | -4 | $(-3, 3)$ |
| 6 | -4 | 3 | 8  | $(-4, 3)$ |
| 7 | -5 | 2 | 4  | $(-5, 2)$ |



3.a)

Ans- Same as Set-A

for clock-wise rotation, sign of angle is changed.

$$\sin(-\theta) = -\sin\theta$$

$$\cos(-\theta) = \cos\theta$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

3.6) Clockwise of set A.

$A(2, 5), B(7, 10) \& C(10, 2)$ .

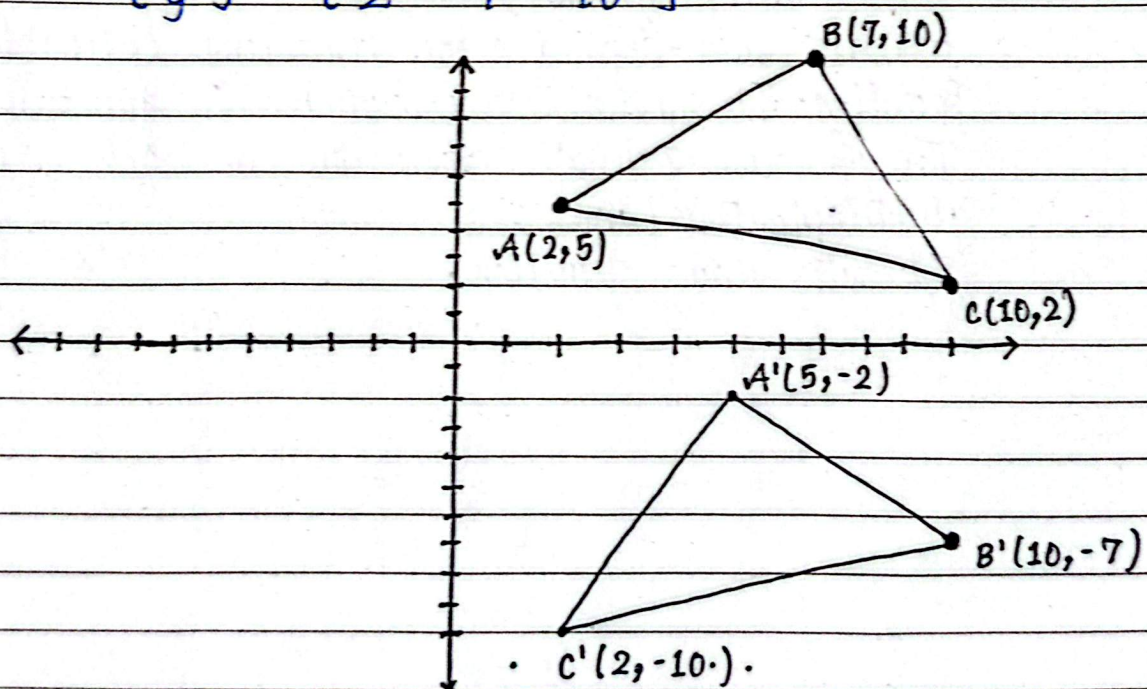
90 about origin.

Solution,

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 2 & 7 & 10 \\ 5 & 10 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 2 & 7 & 10 \\ 5 & 10 & 2 \end{bmatrix}$$

$$\therefore \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 5 & 10 & 2 \\ -2 & -7 & -10 \end{bmatrix}$$



4. Write short notes on:

a) `glutInit()` and `glutDisplayFunc()`

`glutInit(<argc>, <argv>)`

→ It initializes graphics mode so that scene can be rendered. It takes two parameters from main function.  
`glutInit(&argc, argv);`

The commands containing parameters are for command line argument. It must be called before any other glut routine.

`glutDisplayFunc()`

→ It calls user-defined scene rendering callback function. It means its parameter is function name.  
The function generally draws on the screen window.  
`glutDisplayFunc(draw);`

It calls draw function.